

Auteur: Siebe Mees
 Studierichting: Informatica
 Oefeningenreeks 4A nr. 60.10

$$\begin{aligned}
 & \int \frac{x^{\frac{3}{4}} + 1}{x^{\frac{1}{2}} + 1} dx \\
 & t = x^{\frac{1}{4}} \left(dt = \frac{1}{4} x^{-\frac{1}{4}} dx = \frac{1}{4x^{\frac{3}{4}}} dx, \quad dx = 4x^{\frac{3}{4}} dt = 4t^3 dt \right) \\
 & \int \frac{t^3 + 1}{t^2 + 1} \cdot 4t^3 dt \\
 & = 4 \int \frac{t^6 + t^3}{t^2 + 1} dt \\
 & = 4 \left(\int \frac{t^6}{t^2 + 1} dt + \int \frac{t^3}{t^2 + 1} dt \right) \\
 & \begin{cases} \frac{t^6}{t^2 + 1} = t^4 - t^2 + 1 - \frac{1}{t^2 + 1} \\ \frac{t^3}{t^2 + 1} = t - \frac{t}{t^2 + 1} \end{cases} \\
 & = 4 \int \left(t^4 - t^2 + 1 + t - \frac{1}{t^2 + 1} - \frac{t}{t^2 + 1} \right) dt \\
 & = 4 \left(\int t^4 dt - \int t^2 dt + \int 1 dt + \int t dt - \int \frac{1}{t^2 + 1} dt - \int \frac{t}{t^2 + 1} dt \right) \\
 & = 4 \left(\frac{t^5}{5} - \frac{t^3}{3} + t + \frac{t^2}{2} - \text{Bgtan}(t) - \frac{1}{2} \ln(t^2 + 1) \right) + C \\
 & = \frac{4t^5}{5} - \frac{4t^3}{3} + 4t + 2t^2 - 4\text{Bgtan}(t) - 2\ln(t^2 + 1) + C \\
 & t = x^{\frac{1}{4}} \\
 & = \frac{4}{5} x^{\frac{5}{4}} - \frac{4}{3} x^{\frac{3}{4}} + 4x^{\frac{1}{4}} + 2x^{\frac{1}{2}} - 4\text{Bgtan}(x^{\frac{1}{4}}) - 2\ln(x^{\frac{1}{2}} + 1) + C
 \end{aligned}$$

References